Advanced Applied Statistics

Lecture 3: Parametric Models, Fisher Information, Exponential Family

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Learning Objectives for this Lecture

- Relate univariate and multivariate families of distributions
- Reflect multivariate normal distribution and its properties
- Explain the role of Fisher information and estimate and apply it in real scenarios
- Explain the multinomial distribution
- Explain why exponential family is favourable model in statistics and recognise fundamental models from the family

Outline

Univariate Families

Multivariate Gaussians

Fisher Information

The Multinomial Distribution

Exponential Family

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Single variable parametric models

Suppose a random variable X takes values in $\mathcal{X}$. We will consider processes described by a parametrized probability density function

$$\mathcal{F} = \{f_\mu(x); x \in \mathcal{X}, \mu \in \Omega\}, \tag{2}$$

Recall that

- Positivity: $f_\mu(x) \geq 0$ for all $x \in \mathcal{X}$ and $\mu \in \Omega$.
- Probability: $\int f_\mu(x)dx = 1$ for all $\mu \in \Omega$

Univariate Families

Definition

A parameter family of probability densistion, or probabilty mass functions, is called **univariate parameter family** if and only the sample space $\mathcal{X}$ is a subset of the real line $\mathbb{R}$.

Familiar univariate families

Table 5.1 *Five familiar univariate densities, and their sample spaces $\mathcal{X}$, parameter spaces Ω, and expectations and variances; chi-squared distribution with n degrees of freedom is* $2\,\mathrm{Gam}(n/2, 1)$.

Name, Notation	Density	$\mathcal{X}$	Ω	Expectation, Variance
Normal $\mathcal{N}(\mu, \sigma^2)$	$\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$	$\mathcal{R}^1$	$\mu \in \mathcal{R}^1$ $\sigma^2 > 0$	μ σ^2
Poisson $\mathrm{Poi}(\mu)$	$\frac{e^{-\mu}\mu^x}{x!}$	$\{0, 1, \ldots\}$	$\mu > 0$	μ μ
Binomial $\mathrm{Bi}(n, \pi)$	$\frac{n!}{x!(n-x)!}\pi^x(1-\pi)^{n-x}$	$\{0, 1, \ldots, n\}$	$0 < \pi < 1$	$n\pi$ $n\pi(1-\pi)$
Gamma $\mathrm{Gam}(\nu, \sigma)$	$\frac{x^{\nu-1}e^{-x/\sigma}}{\sigma^\nu \Gamma(\nu)}$	$x \geq 0$	$\nu > 0$ $\sigma > 0$	$\sigma\nu$ $\sigma^2\nu$
Beta $\mathrm{Be}(\nu_1, \nu_2)$	$\frac{\Gamma(\nu_1+\nu_2)}{\Gamma(\nu_1)\Gamma(\nu_2)} x^{\nu_1-1}(1-x)^{\nu_2-1}$	$0 \leq x \leq 1$	$\nu_1 > 0$ $\nu_2 > 0$	$\nu_1/(\nu_1+\nu_2)$ $\frac{\nu_1\nu_2}{(\nu_1+\nu_2)^2(\nu_1+\nu_2+1)}$

Familiar univariate families

Exercise:

1. Show that $\sum_{x=0}^{\infty} f_\mu(x) = 1$.
2. Show that $\mathbb{E}[x] = \mu$
3. Show that $\mathrm{Var}[x] = \mu$.

for the Poisson distribution.

- The parametrized families are constructed so that they are easy to manipulate.

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Gaussian n-Variate Random Variable

From Kaipio & Somersalo 2005, pp. 72–76.

Definition

Definition 3.2. *Let $x_0 \in \mathbb{R}^n$ and $\Gamma \in \mathbb{R}^{n\times n}$ be a symmetric positive definite matrix, denoted by $\Gamma > 0$ in the sequel. A Gaussian n-variate random variable X with mean x_0 and covariance Γ is a random variable with the probability density*

$$\pi(x) = \left(\frac{1}{2\pi|\Gamma|}\right)^{n/2} \exp\left(-\frac{1}{2}(x-x_0)^{\mathrm{T}}\Gamma^{-1}(x-x_0)\right),$$

where $|\Gamma| = \det(\Gamma)$. In such case, we use the notation

$$X \sim \mathcal{N}(x_0, \Gamma).$$

Some Elementary Matrix Properties: Schur Complements

Definition

Definition 3.3. *Let*

$$\Gamma = \begin{bmatrix} \Gamma_{11} & \Gamma_{12} \\ \Gamma_{21} & \Gamma_{22} \end{bmatrix} \in \mathbb{R}^{n \times n}$$

be a positive definite symmetric matrix, where $\Gamma_{11} \in \mathbb{R}^{k \times k}$, $\Gamma_{22} \in \mathbb{R}^{(n-k)\times(n-k)}$, $k < n$, *and* $\Gamma_{21} = \Gamma_{12}^{\mathrm{T}}$. *We define the* Schur complements $\tilde{\Gamma}_{jj}$ *of* Γ_{jj}, $j = 1, 2$, *by the formulas*

$$\tilde{\Gamma}_{22} = \Gamma_{11} - \Gamma_{12}\Gamma_{22}^{-1}\Gamma_{21}, \quad \tilde{\Gamma}_{11} = \Gamma_{22} - \Gamma_{21}\Gamma_{11}^{-1}\Gamma_{12}.$$

- Note: positive definitess of Γ implies that the $\Gamma_{jj}, \quad j = 1, 2$ are also positive definite.

Some Elementary Matrix Properties: Schur Complements (2)

Lemma

Lemma 3.4. *Let Γ be a matrix satisfying the assumptions of Definition 3.3. Then the Schur complements $\tilde{\Gamma}_{jj}$ are invertible matrices and*

$$\Gamma^{-1} = \begin{bmatrix} \tilde{\Gamma}_{22}^{-1} & -\tilde{\Gamma}_{22}^{-1}\Gamma_{12}\Gamma_{22}^{-1} \\ -\tilde{\Gamma}_{11}^{-1}\Gamma_{21}\Gamma_{11}^{-1} & \tilde{\Gamma}_{11}^{-1} \end{bmatrix}. \tag{3.8}$$

- It also follows that $\det(\Gamma) = \det(\Gamma_{11})\det(\tilde{\Gamma}_{11})$
- Due to symmetry of Γ, we have $\Gamma_{22}^{-1}\Gamma_{21}\Gamma_{22}^{-1} = \Gamma_{11}^{-1}\Gamma_{21}\Gamma_{11}^{-1}$.

Conditional Distributions of the n-Variate Gaussians

Theorem

Theorem 3.5. *Let $X : \Omega \to \mathbb{R}^n$ and $Y : \Omega \to \mathbb{R}^k$ be two Gaussian random variables whose joint probability density $\pi : \mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}_+$ is of the form*

$$\pi(x, y) \propto \exp\left(-\frac{1}{2}\begin{bmatrix} x - x_0 \\ y - y_0 \end{bmatrix}^{\mathrm{T}} \begin{bmatrix} \Gamma_{11} & \Gamma_{12} \\ \Gamma_{21} & \Gamma_{22} \end{bmatrix}^{-1} \begin{bmatrix} x - x_0 \\ y - y_0 \end{bmatrix}\right). \tag{3.11}$$

Then the probability distribution of X conditioned on $Y = y$, $\pi(x \mid y) : \mathbb{R}^n \to \mathbb{R}_+$ is of the form

$$\pi(x \mid y) \propto \exp\left(-\frac{1}{2}(x - \overline{x})^{\mathrm{T}} \tilde{\Gamma}_{22}^{-1} (x - \overline{x})\right),$$

where

$$\overline{x} = x_0 + \Gamma_{12}\Gamma_{22}^{-1}(y - y_0).$$

Marginal Distributions of the n-Variate Gaussians

Theorem

Theorem 3.6. *Let X and Y be Gaussian random variables with joint probability density given by (3.11). Then the marginal density of X is*

$$\pi(x) = \int_{\mathbb{R}^k} \pi(x,y)dy \propto \exp\left(-\frac{1}{2}(x-x_0)^{\mathrm{T}}\Gamma_{11}^{-1}(x-x_0)\right).$$

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Recap: Desired properties for an Estimator

- **Easy to compute**
- **Unbiased**: $\mathbb{E}[T] = \theta$
- **Efficient**: $\mathrm{Var}(T)$ is as low as possible
- **Invariant to transformation** of the parameter θ: if the distribution was parametrised in terms of $g(\theta)$ then $g(T)$ would be an estimator for $g(\theta)$

Exercise

Which properties does the MLE have?

Further property of the MLE

Theorem

Consistency The maximum likelihood estimator is **consistent**, *i.e., the MLE* $\hat{\theta}$ *converges to the true parameters* $\theta_0 \in \Theta$ *in probability, or,*

$$\hat{\theta} \xrightarrow{\mathrm{P}} \theta_0. \tag{1}$$

We can say something more about the **asymptotic distribution** of the MLE.
To do that we need to introduce one more new concept: **Fisher Information**.

Fisher Information

Definition

Let functional $l(\theta)$ denote the log likelihood $\log L(x;\theta)$, where the parameter vector is θ. The gradient of the log likelihood $\nabla_\theta l(\theta) = \nabla_\theta \log L(x;\theta)$ is called the **score**.

Note: for the MLE, the expected value of the score is zero (why?), i.e., $\mathrm{E}\{\nabla_\theta l(\theta)\} = 0$.

Definition

Fisher information $\mathcal{I}(\theta)$ is the covariance matrix of the score or

$$\mathcal{I}(\theta) = \left\{ \int \left(\frac{\partial}{\partial \theta_i} \log L(x;\theta) \right) \left(\frac{\partial}{\partial \theta_j} \log L(x;\theta) \right) L(x;\theta) \mathrm{d}x \right\}. \tag{2}$$

Fisher Information (2)

Theorem

If the log likelihood is twice differentiable, under certain regularity conditions,

$$\mathcal{I}(\theta) = -\mathrm{E}\left\{\frac{\partial^2}{\partial\theta_i\theta_j}\log L(X;\theta)\right\} \tag{3}$$

In other words, Fisher information equals the **Hessian of the expected negative log likelihood**.

Further properties of the MLE (2)

The maximum likelihood estimator is **asymptotically normal**.

Theorem

Assume that $\hat{\theta}_0 \in \Theta$ in probability. Under certain regularity conditions,

$$\hat{\theta} \xrightarrow{\mathrm{d}} \theta \sim N(\theta_0, \mathcal{I}(\theta_0)^{-1}). \tag{4}$$

- The normal distribution has the centre at θ_0
- Inverse of the Fisher information is the covariance matrix of the normal distribution.

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The Multinomial distribution

Given a probability vector $\pi \in \mathbb{R}_+^L$, and a real number n, the multinomial distribution, denoted $\mathrm{Mult}_L(n, \pi)$ is given by:

$$f_\pi(x) = \frac{n!}{x_1!...x_L!} \prod_{l=1}^{L} \pi_l^{x_l} \tag{4}$$

- Try to convince yourself that $f_\pi(x)$ is a probability distribution.
- There is a close relation between the Multinomial distribution and the Poisson distribution.

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Example

Example

Let us consider the Poisson family $\mathrm{Poi}(\mu)$. The ratio of two Poisson densities, specified by the parameters μ and μ_0, is

$$\frac{f_\mu(x)}{f_{\mu_0}(x)} = \mathrm{e}^{-(\mu-\mu_0)} \left(\frac{\mu}{\mu_0}\right)^x \quad \Leftrightarrow \quad f_\mu(x) = \mathrm{e}^{\alpha x - \psi(\alpha)} f_{\mu_0}(x), \tag{5}$$

where $\alpha = \log(\mu/\mu_0))$ and $\psi(\alpha) = \mu_0(\mathrm{e}^\alpha - 1)$.

- α is a **natural** or **canonical parameter**.
- $y = t(x)$ is the **sufficient statistic**.
- The Poisson is a **one-parameter family** since α and x are scalars.
- $\psi(\alpha)$ is a normalising function.

Exponential Family: General Definition

Definition

Let α be a real parameter vector. A family of distributions is said to belong to the the **exponential family** if the probability density function (or probability mass function for discrete distributions), can be written in the form

$$f_\alpha(\mathbf{x}) = h(\mathbf{x}) \exp\left(\eta(\alpha)^{\mathrm{T}} \mathbf{t}\left(\mathbf{x}\right) - \psi\left(\alpha\right)\right) \tag{6}$$

for **natural parameter vector** η, and for given functions $\mathbf{t}$ and h. The statistic $\mathbf{t}(\mathbf{X})$ is referred to as a **sufficient statistic**. The function

$$\psi(\alpha) = \log\left(\int_X h(\mathbf{x}) \exp(\eta(\alpha)^{\mathrm{T}} \mathbf{t}(\mathbf{x})) \mathrm{d}\mathbf{x}\right) \tag{7}$$

is called the normalising function[1].

[1] Also known as the the **cumulant generating function** or **log-partition function**.

Cumulant Generating Function

- The normaliser $\psi(\alpha)$ is also called the cumulant generating function, since its derivatives yields the cumulants of $\mathbf{y} = \mathbf{t}(\mathbf{x})$
- For instance, $\nabla\psi(\alpha) = \mathrm{E}_\alpha\{\mathbf{y}\} \equiv \beta$ and $\left(\frac{\partial^2\psi(\alpha)}{\partial\alpha_j\partial\alpha_k}\right) = \mathrm{Cov}\{\mathbf{y}\}$

Examples

Example

Normal distribution: unknown mean, known variance The probability density function

$$f_\sigma(x;\mu) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/2\sigma^2} \tag{8}$$

This is a single-parameter exponential family with

$$h_\sigma(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{x^2/(2\sigma^2)} \tag{9}$$

$$t_\sigma(x) = \frac{x}{\sigma} \tag{10}$$

$$\psi_\sigma(\mu) = \frac{\mu^2}{2\sigma^2} \tag{11}$$

$$\eta_\sigma(\mu) = \frac{\mu}{\sigma} \tag{12}$$

Examples

Example

Normal distribution: unknown mean, unknown variance The probability density function

$$f(x;\mu,\sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/2\sigma^2} \tag{13}$$

This is a two-parameter exponential family with

$$h(x) = \frac{1}{\sqrt{2\pi}} \tag{14}$$

$$\mathbf{t}(x) = (x, x^2) \tag{15}$$

$$\psi(\mu) = \frac{\mu^2}{2\sigma^2} + \log|\sigma| = -\frac{\eta_1^2}{4\eta_2} + \frac{1}{2}\log\left|\frac{1}{2\eta_2}\right| \tag{16}$$

$$\eta(\mu) = \left(\frac{\mu}{\sigma^2}, -\frac{1}{2\sigma^2}\right) \tag{17}$$

Properties

Exponential family of distributions are extremely useful for statistical analysis. Often only the exponential family has these properties. Examples:

- Tthe only families with sufficient statistics that can summarize arbitrary amounts of independent identically distributed data using a fixed number of values. (Pitman–Koopman–Darmois theorem)
- Conjugate priors, an important property in Bayesian statistics.
- The posterior predictive distribution of a random variable with a conjugate prior can always be written in closed form.
- In the mean-field approximation in variational Bayes (Bayesian networks), the best approximating posterior distribution of an exponential-family node with a conjugate prior is in the same family as the node.

Quiz